

Experiment 2

AIM: Study of LM35 IC temperature sensor.

Apparatus Required:

- Temperature Transducer Trainer (MI-IN04).
- Patch Cords
- Power cable
- Digital Multimeter(DMM)

Theory:

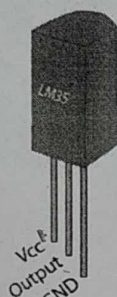


Fig2.1: LM35-IC temperature sensor

LM35 is a precision IC temperature sensor with its output proportional to the temperature (in °C). The sensor circuitry is sealed and therefore it is not subjected to oxidation and other processes. With LM35, temperature can be measured more accurately than with a thermistor. It also possess low self-heating and does not cause more than 0.1 °C temperature rise in still air. The operating temperature range is from -55°C to 150°C. The output voltage varies by 10mV in response to every °C rise/fall in ambient temperature, i.e., its scale factor is 0.01V/°C.

Safety Precaution:

- Make all the connection before power on the trainer.
- All connection should be tight.
- The circuit should be off while changing the connections.
- Switch off the supply of trainer and remove all connections after completing the experiment.
- Push Button Switch of Heating Element Control System must be set on External(push-out).

Procedure:-

1. Make Connection Diagram as shown below figure2.2.

Connection Description

- LM35 output connected to Temperature Display(+) and also Digital Multimeter. (+).
 - Temperature Display (-) and Digital Multimeter(-) connected to Ground.
 - Heater Input Connected to Heating control system (To Heater).
2. Set Digital Multimeter in mV Range.
 3. Switch on trainer and note down reading of room temperature, showing by temperature display and value of Voltage measured by digital multimeter.
 4. Set Heating Control System unit to manual mode by switching push-button to "push out" condition.
 5. As we see that temperature is increasing gradually and Output Voltage of LM35 is also increasing as a function of temperature.
 6. Note down Reading of LM35 Output Voltage and Temperature on different interval.
 7. Now plot the graph between Temperature and resistance.

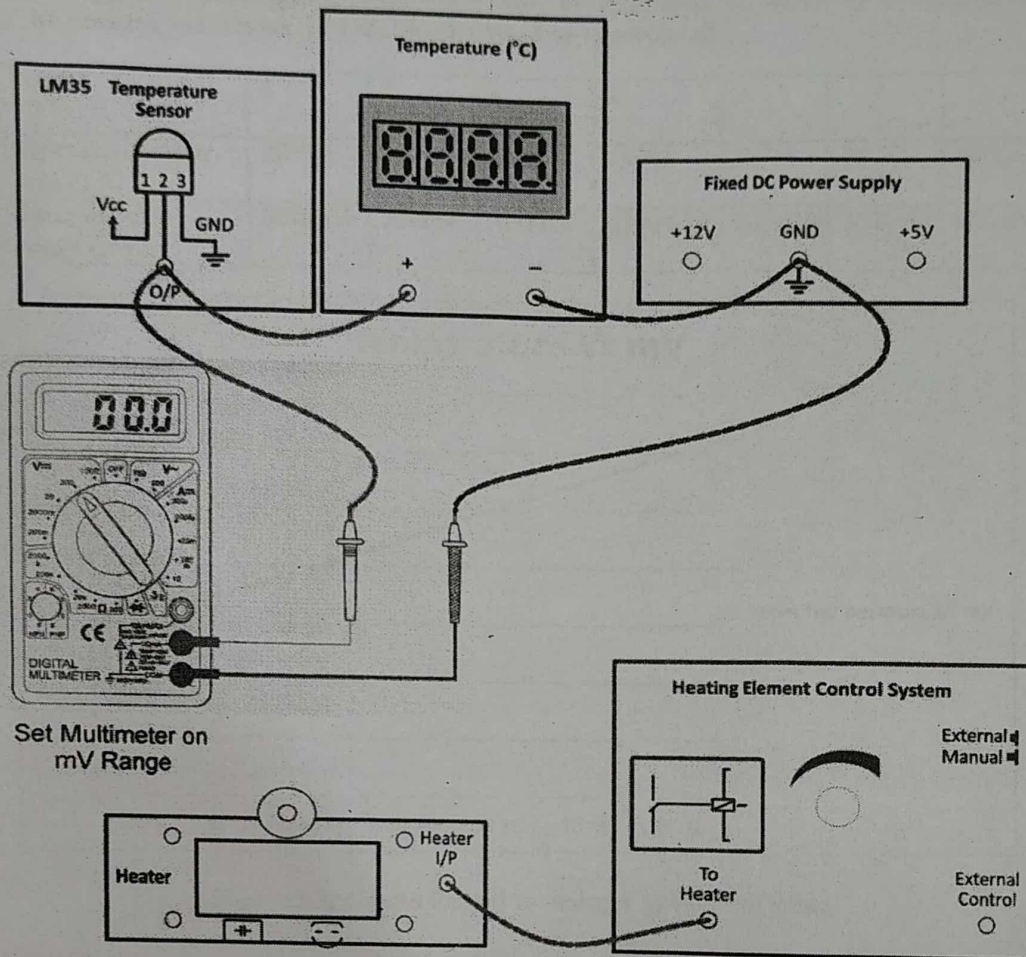


Fig2.2: Connection circuit of LM35 Temperature sensor

Observation Table:

S. No.	1	2	3	4	5	6	7
Temperature (From Display) °C							
LM35 Output Voltage (mV)							

Expected Readings:

Note: Expected Reading and observation table reading may be varied by $\pm 5\%$, depend on the accuracy of sensor and multimeter used in experiment.

S. No.	1	2	3	4			
Temperature (From Display) °C	30 °C	35 °C	40 °C	45 °C	50 °C	55 °C	60 °C
LM35 Output Voltage (mV)	300mV	350mV	400mV	450mV	501mV	552mV	601mV

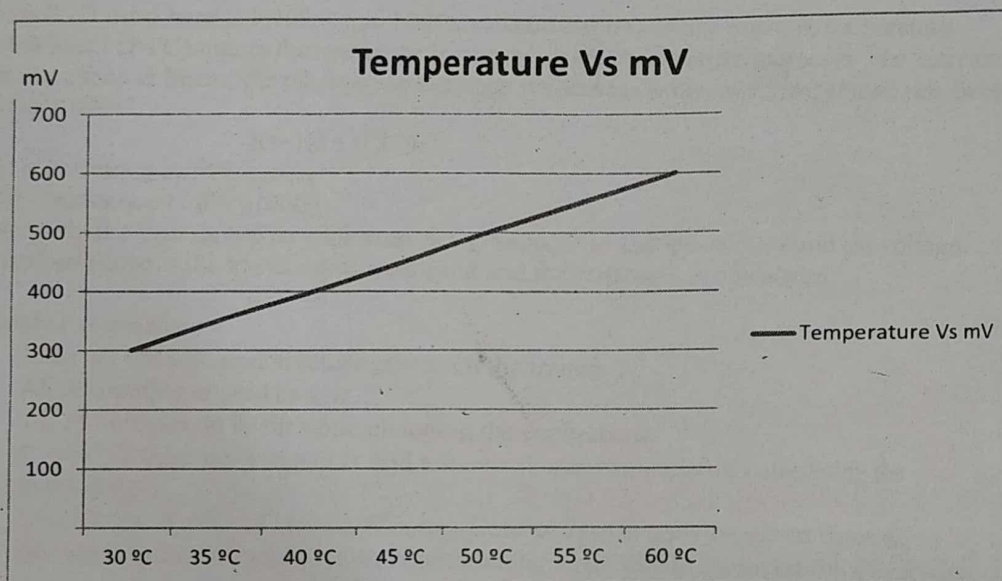


Figure2.3: Graph Plotted by using Expected Reading

Conclusion:

Here we studied temperature increasing then voltage is also increasing linearly in LM35 temperature sensor.

Experiment 4

AIM: To study the characteristics of PT100 RTD.

Apparatus Required:

- **Temperature Transducer Trainer (MI-IN04).**
- Patch Cords
- Power cable
- Digital Multimeter(DMM)

Theory:

The RTD used here is Pt100; it has 100Ω resistance at 0°C. It has positive temperature coefficient (PTC) means the resistance increases as the temperature increases. The increases in resistance is linear, the relationship between resistance change and temperature rise being 0.00385Ω/°C.

$$R_t = R_0 + 0.385t$$

R_t = resistance at temperature °C.

R_0 = resistance at 0°C (100Ω).

Normally the unit would be connected to DC supply via a series resistor and the voltage developed across the transducer is measured and the resistance is calculated.

Safety Precaution:

- Make all the connection before power on the trainer.
- All connection should be tight.
- The circuit should be off while changing the connections.
- Switch off the supply of trainer and remove all connections after completing the experiment.
- Push Button Switch of Heating Element Control System must be set on External.
- Heating Element Control System Potentiometer Knob should be set on full Clockwise.

Procedure:-

1. Make Connection Diagram as shown below figure4.1.

Connection Description

- RTD PT100 output (A & B) Connected to Digital Multimeter.
 - Heater Input Connected to Heating control system (To Heater).
 - LM35 Connection same as previous Experiment No. 2 for reading temperature.
 - PT100 Sensor having 3 wires are connected to Platinum RTD Block(A,A & B).
2. Set Digital Multimeter in 200Ω Range.
 3. Insert Sensor into heating tube.
 4. Switch on trainer and note down reading of room temperature, showing by temperature display and value of resistance measured by digital multimeter.
 5. Set Heating Control System unit to manual mode by switching push button on push out condition.
 6. As we see that temperature is increasing gradually and Resistance of PTC is increasing.
 7. Note down Reading of Resistance and Temperature on different interval.
 8. Now plot the graph between Temperature and resistance.

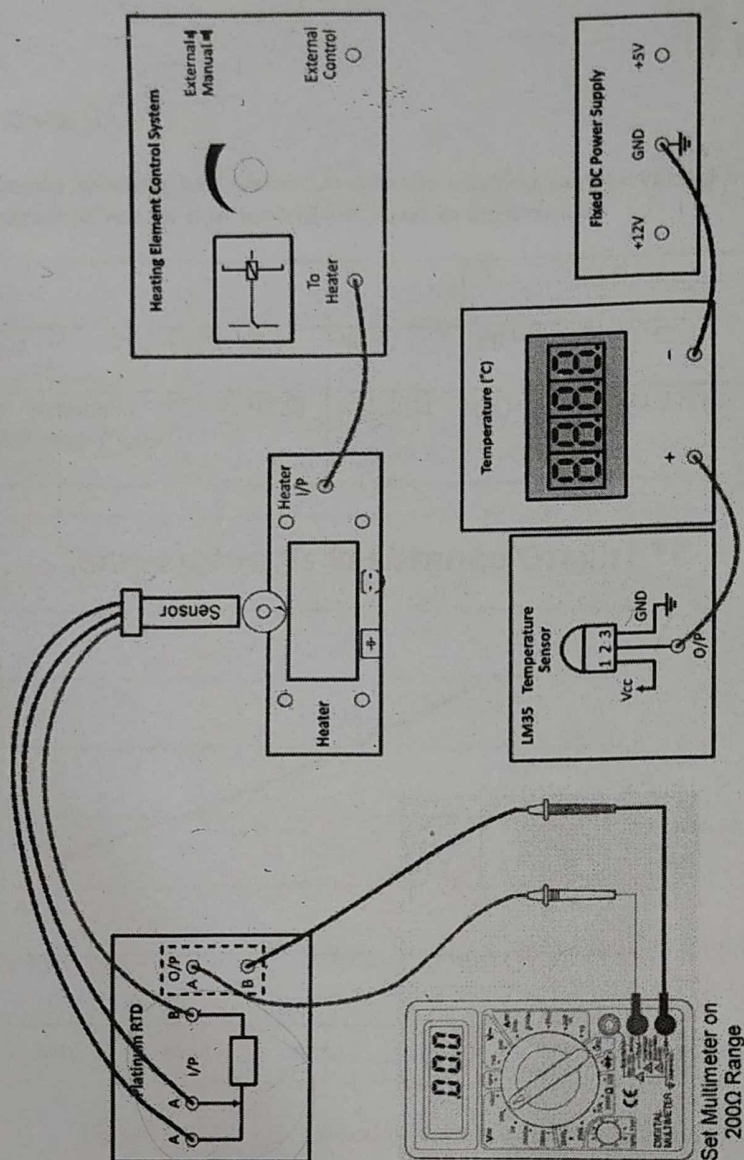


Fig4.1: Connection circuit of PT100 RTD

Observation Table:

S. No.	1	2	3	4	5
Temperature (From Temperature Display) °C					
Thermistor Resistance (From digital multimeter) KΩ					

Expected Readings:

Note: Expected Reading and observation table reading may be varied by $\pm 5\%$, depend on the accuracy of sensor and multimeter used in experiment.

S. No.	1	2	3	4	5
Temperature (From Temperature Display) °C	30 °C	40 °C	50 °C	60 °C	70 °C
Thermistor Resistance (From digital multimeter) K Ω	111.97 Ω	115.23 Ω	119.01 Ω	123.47 Ω	128.52 Ω

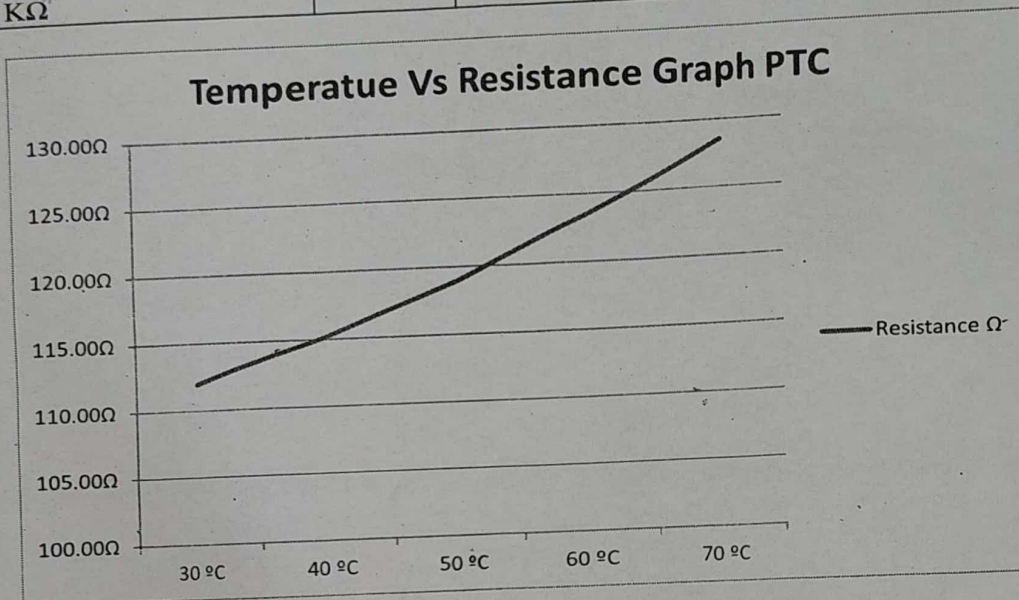


Figure3.2: Graph Plotted by using Expected Reading

Conclusion: - Here we studied that resistance of PT100 RTD starts increasing as temperature starts rising/increasing.